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Sustainable Logistics Network Optimization From Dry Ports to Seaport: A Case Study From Pakistan

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Abstract

Several considerations are explored in each mode of transportation for the safe, quick, and environmentally friendly movement of goods. To obtain the advantages in all transportation modes, it is critical to develop appropriate models and algorithms. By expanding our understanding of the planning of bi-objective optimization problems, this study contributes to the field of real-time green logistics analysis. The paper provides case studies based on road and road-rail transportation from dry ports to a seaport in Pakistan. Analysis of the cases shapes the economic and environmental planning of transport. Because there is little research on this topic in Pakistan, the innovative aspect is to reduce the cost and greenhouse gas (GHG) emissions of road and intermodal freight transportation between dry ports and seaport. A mixed-integer linear programming model is used for the formulation of the bi-objective problem. Pareto front solutions are provided to trade-off the cost and GHG emissions in delivering goods from dry ports to seaport in Pakistan. The study uses real-life data from one of the largest logistics service providers in Pakistan. Results show that intermodal transportation is 82.6% more economical and 78% greener compared with road freight transportation. Further, this study could be extended by including other factors such as noise, air pollution, and traffic accidents.

Keywords

rail, freight rail transportation, freight, intermodal, railroads, sustainability and resilience, transportation and sustainability, air quality and greenhouse gas mitigation, air quality modeling, emissions and air quality management, emissions modeling

The debate on freight transportation and logistics increasingly includes the challenge of sustainable development. The vital contributors in green logistics development are the mitigation of adverse environmental effects and the realization of modal shift in transportation networks to improve cost and quality efficiencies. The introduction of a range of measures is needed to pursue these goals, and it is more effective to combine all the actions, policies, and initiatives into an integrated package of measures than to deal with them individually. The inter-port model has been introduced to integrate the cost and sustainability of freight distribution to and from dry ports using containers. This model examines all the effects of inland container flow and its possible external and internal costs. Data analysis shows the impact of operational measures and policies on dry port multimodal distribution (*1*). Container transportation is proliferating globally, and it is forecast that this growth will increase in the future. To expand their share of the market in a competitive environment, companies are investing in containerization. A new port forecasting approach was presented by Zondag et al. (*2*). The model follows the logistics chain and estimates the impacts of pricing, infrastructure, and policy-making on the port itself and its connection with dry ports. The ports of Antwerp, Rotterdam, Bremen, and Hamburg are used to illustrate this model (*2*). A study conducted in Malaysia adopts a

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qualitative approach to evaluate the connectivity of seaports and dry ports from their physical properties. The study results show that the majority of the seaports in Malaysia are connected with dry ports by both rail and road. However, they are most strongly linked with highways, and the quality of the dry port–seaport connections can be improved (*3*). The chains of transportation for dry ports are jointly determined by dry ports, seaports, exporters, importers, and intermodal carriers. Talley and Ng (*4*) proposed a model to design the joint choice problem. The study found that the effects on dry port transportation chain choices of dry ports and seaports' throughput, intermodal carriers' profit, and cost of exporters' and importers' logistics can either be extenuated or amplified, depending on the existence of indirect effects (*4*).

Road transportation is the most prominent mode of transportation between dry ports and seaports. The pricing of road transportation has an impact on the competitiveness of the ports. Meersman et al. analyzed the effect of cost function on various actors involved in the logistics chains of ports (*5*). The cost of road freight transportation varies from country to country based on variation in costs of fuel, taxation, wages, maintenance, tolls, repairs, tires, parking spaces, and so forth. In Kotsios and Folinas (*6*), the cost of road freight transportation is measured and evaluated for 20 countries in Europe. The cost is divided into four main categories: drivers' wages, fuel consumption, tolls, and tires. The results of this study show a significant variation in costs between the countries. Poland and Bulgaria are the countries with the lowest road freight transportation costs, while Austria, the UK, and Norway are the highest cost countries. The most significant difference is observed in the cost of toll taxes, followed by wages of drivers, fuel consumption, and tires (*6*). Road transportation for dry port services can be challenged by using different modes of transportation like road-rail or sea-road because of environmental, cost, and congestion constraints. However, combined transportation must compete with the advantages of road transportation because road and combined transportation are not directly comparable given the differences in physical services. In Frémont and Franc (*7*), the case of dry port services between the Paris region and the seaport of Le Havre is investigated for road and combined transportation. This investigation shows that the competitiveness of combined transportation with regard to cost varies significantly compared with road transportation (*7*).

With the flexibility provided by road transportation, multimodal and intermodal container shipments are a growing industry for U.S. soybean exports (*8*). To understand the best approach for increasing U.S. economic competitiveness in the emerging sector, Bai et al. produced a detailed model and cost analysis focused on U.S. soya container exports. The model calculates the supply chain costs of alternative shipping routes from a domestic production site to a foreign port, using mode-specific transportation network and cost information. The study calculated the time, size, and transport cost for each modal section for individual candidate routes. This cost analysis model is part of a larger research project to formulate strategies for improving networks and transactions in the light of current and future developments in transport sectors and the global economy (*8*). The CTRail costing model is designed to supersede the less flexible uniform model of rail costing by providing policymakers and freight stakeholders with a tool to perform independent studies on rail freight movement. CTRail is based on a simulation of the horsepower needed to transport a given tonne of cargo over a specified distance at a specified speed. The amount of fuel used is calculated with the horsepower measured and transformed into fuel costs and emissions. It is thought that the cost of maintenance will vary with the number of devices (locomotives and railways) and the distance traveled. Crew pay is calculated as an hourly rate of input based on a regular rate based on time and expenditure for traveling. Multiple user-specified entries are also available in this model. However, the current version of CTRail limits the types of analysis that can be done, despite the versatility of the analytical routes and corridors (*9*).

Logistics designs are driven by the need to reduce the cost of transport but also to improve customer service. Consequently, in recent years, the emphasis on environmental issues is increasing in transportation and logistics services. Harris et al. (*10*) assessed the optimized traditional cost approach's impact on overall CO_2 emissions in logistics by studying the supply chain structure. The findings of the investigation indicate that a cost-optimized design does not always translate to the best CO_2 emission solution. However, only one case study was investigated, and the assumptions about the transportation data also limit the study because a variety of vehicles were used in the supply chain to transport the commodities (*10*).

The demand for transport services is continuously increasing and is expected to grow in the future. This increase in transportation activities is increasing the CO_2 emissions in the air. The major part of such CO_2 emissions come from road transportation. Effective policies and strategies can reduce the growth of CO_2 emissions. Under the constraints of fuel and travel demands, Mustapa and Bekhet designed an optimization model to estimate CO_2 emissions for road transportation in Malaysia (*11*). Many strategies of mitigation have been used to analyze the potential effects of CO_2 emissions. According to that study, reducing the fuel price subsidy will save up to 652,000 tonnes of fuel and reduce CO_2 emissions by 5.55%. However, policymakers have to face the difficulty of balancing political and economic efficiency

and social acceptability with protecting the environment. Therefore, sharing information among prosperous countries on new technologies and mitigation strategies is essential to minimize the gap between government policies and the market's response. One possible way of decreasing CO_2 emissions is to use intermodal or multimodal transportation. The impact of the dry port is investigated in a study conducted in Finland by differentiating two different configurations (*12*). The shippers move their products directly to the seaport in the first one while they use the dry ports in the second. Discrete-event simulation is used to evaluate the systems. The objectives of this study are to minimize the cost of transportation and CO_2 emissions. Different scenarios are used to estimate the future cost and CO_2 emissions. By implementing the dry port network in the context of Finland's transportation network, the results show that CO_2 emissions can decrease by 45% with less than 1% increase in the total transportation cost. However, no information concerning transport movements in Finland is available, and capacity limitation is not used in the model (*12*).

The transition to more effective and less polluting modes may be a crucial policy priority for achieving energy usage and carbon reduction objectives in the transport sector. However, it is not always easy to make distinctions between modes. Data on average energy and emissions, especially for rail, are frequently used, but they may not be relevant in a given situation. Some UK train operators have recently equipped their trains with electricity meters from which data are collected for energy usage (*13*). This makes it possible to analyze how various conditions, including the train type and operation type, affect energy use and associated pollution. Comparisons between road and rail are rendered for current records. While more precise data would help advise policy and certain decision making, the average data still plays a significant role in the overall picture (*13*).

The driving force behind sustainable infrastructure is transport sustainability. As economies develop, the policy of reductions in greenhouse gas (GHG) emissions cannot ignore the increased importance of transport. In this context, assessing the relationship between CO_2 pollution from transportation and economic development is critical for developing appropriate transportation policies. Sousa et al. (*14*) examined the relationship developed in this area for the Portuguese transport sector using a non-linear co-integration methodology. The key inference is the growing monotonous association between gross domestic product and CO_2 emissions from transport, meaning that economic development in itself is insufficient to offset emissions. Therefore, it is necessary to formulate realistic public policy that can affect both passengers and freight mobility strategies to mitigate CO_2 emissions without jeopardizing economic growth (*14*).

Methodologies for computation of green routing and GHG emissions are significant, in addition to the impact of transportation on GHG emissions. The vehicle routing problem (VRP) has been studied extensively in the literature. These studies take into account a wide range of restrictions and objectives, and they use a variety of modeling and solution approaches. Recent research (*15*) has designed the special VRP with environmental concerns using flow-based mixed-integer linear programming to provide an exact solution for school bus routing problems involving homogeneous and heterogeneous vehicle fleets. When compared with homogeneous fleets, heterogeneous fleets save 30% on total costs. That research also looks at the trade-offs between trip durations, journey times, and fleet component costs (*15*).

A homogenous fleet of ships (of the same type, size, and cost) transports full shipload commodities from loading ports to discharging ports. This means that a ship may only carry one cargo at a time, and each cargo is transported directly from its loading port to its matching unloading port (*16*). An enormous challenge in optimization is the problem of vehicle routing when the fleet is heterogeneous. Since most organizations need to transport a variety of products, they typically maintain a diverse fleet of vehicles. Vehicles with varied capacity and costs must be routed in a precise way in a heterogeneous fleet vehicle routing problem (HVRP). The HVRP plans vehicle routes so that each customer is only served once, demand does not surpass the capacity of the vehicle assigned to it, and total costs are maintained at a minimum (*17*, *18*). In Hulagu et al. (*19*), a mixed-integer linear programming model is used to address the routing problem for a university's electric shuttle service. Instead of using a heuristic technique, an explicit solution method is presented in the case study. According to the results of the study, 18.25 kg of CO_2 emissions could be saved per day when compared with the current shuttle service plan (*19*).

Solving a multi-objective problem involves dealing with two unique sorts of problems: searching and making decisions (*20*). The first aspect is the sampling of the feasible set for Pareto optimum solutions during the optimization process. Large and complex search spaces, especially in single-objective optimization, can complicate the search and limit the use of exact optimization methods like linear programming (*21*). Choosing an acceptable compromise option from the Pareto-optimal set is the focus of the second part. There must be a human decision-maker to make the sometimes difficult trade-offs between competing objectives. Decision making before exploration, search before decision making, and decision making during exploration are all examples of multi-objective optimization approaches that fall into one of three major groups (*20*). However, the exact

solution method is applied for the solution of vehicle routing problems. Studying the road network, an internal university shuttle fleet, and the time-varying demand of passengers, as well as vehicle dynamics, and battery recharge elements in the formulation of the model are all taken into consideration when generating an ideal routing plan for the shuttle fleet. The study's primary objective is to reduce the cost of vehicle operation, battery recharge, and station purchase. Mixed-integer linear programming models are used to apply the exact solution method. As the study's findings demonstrate, accurate routing systems rely heavily on the real-world road network. However, complex networking, on the other hand, is a problem (*22*).

Several other real-world driving factors, in addition to road slope and altitude, have a substantial impact on vehicle emissions (*23–26*). A significant impact of road grade on vehicle exhausts has been observed in numerous studies; while a vehicle is traveling, a positive road grade (uphill) results in a greater load, whereas a negative road grade (downhill) results in a decreased load (*27–29*). Using a portable emission measurement system (PEMS), the influence of driving styles and road slope on gaseous exhaust emissions is explored in a study. Various driving criteria are selected to characterize distinct driving styles. On the basis of acceleration-based metrics, the severity of travel journeys (severe versus usual) is determined. PEMS calculations of NOx and CO_2 emissions revealed a good correlation with driving parameters. CO_2 and NOx emissions increased by 20 to 40% and 50 to 255%, respectively, when compared with normal travel. The grade of the road is determined using the segment method. There is a linear link between CO_2 and NOx emissions and road slope for all rural, urban, and motorways. The road gradient ranging from 0 to 5% resulted in a 65 to 81% rise in CO_2. Depending on the type of road, NOx increases by 85 to 115%. Because of high load points, steep road gradients produce a substantial amount of pollutants (*30*).

From the concept of Pareto efficiency, Pareto-optimal solutions are originated, which are proposed to study income distribution and economic efficiency (*31*). The Pareto front is defined as a modification to a diverse allocation that makes at least one objective function better off and other objectives worse off, and the distribution is termed "Pareto efficient" or "Pareto optimal" when no further improvement can be made (*21*). There have been numerous applications of the Pareto optimality idea in optimization, which tries simultaneously to maximize multiple objective functions according to particular constraints (*32*). Multi-objective problems can be accurately described by developing a Pareto-compliant ranking method that favors nondominated solutions to recognize Pareto-optimal candidate solutions. A method that adheres to this practice must generate and operate on a pool of potential solutions. Population-based evolutionary approaches such as genetic algorithms, ant colony optimization, and particle swarm optimization have gained widespread recognition and popularity in solving multi-objective optimization problems (*33–43*).

To achieve or satisfy customer needs, modeling the logistics device aims to find the most cost effective or efficient system configuration. Optimization, simulation, or heuristics analytical machine analysis were used in the past to accomplish this. Yet a new generation of strategies has emerged in the last two decades as a viable approach for quickly delivering near-optimal solutions to problems that cannot be solved using precise optimization. New metaheuristic advances are described in a study that examines the potential of these emerging technologies to address a wide range of logistics and supply chain issues (*44*). This study focuses on the optimization of ant colonies, genetic algorithms, simulated annealing, and metaheuristics for tabu search. Methods based on metaheuristics are used to assess supply chain risks and delays, intermodal processes, customer care costs and benefits, backhaul policies, and issues related to the location and travel of vehicles near one another. It concludes by discussing how academic institutions can use these approaches in the classroom to ensure that students entering the logistics and supply chain profession are prepared with up-to-date knowledge of current supply chain technologies (*44*).

Our research provides a bi-objective mixed-integer linear programming (MILP) model for the construction of a green road-intermodal service network based on Demir et al. (*45*). Time-related costs are not included in the model. Our model combines financial and environmental objectives and optimizes them using a genetic algorithm. The research gaps addressed in the study are shown in Table 1.

This article is divided into five sections. The next section includes the problem's basic technical context, description,

Table 1. Research Gaps

Research framework	Demir et al. (*45*)	Our work
Model	MILP	MILP
Solution algorithm	$\in$ -constraint weighting method with normalization (WMN)	Genetic algorithm
Objectives	Bi-objective	Multi-objective
Transport operation	Intermodal	Multimodal
Background of study	Dry port, Europe	Dry ports and seaport, Pakistan

Note: MILP = mixed-integer linear programming.

mathematical model assumptions, and some details on the solution technique. The third section presents the main parameters and initial data for the calculations. The fourth section reveals the findings and discussions on various real-life scenarios. The final section presents the conclusion, limitations, and potential future phases of this study. The study's major contributions are as follows:

- MILP modeling for Pakistani road and road-rail freight delivery networks from dry ports to seaports, taking into account overall transportation costs and GHG emissions.
- Analysis of the cost of a road and road-rail freight distribution network in Pakistan using real-world data.
- To generate Pareto front solutions for decision makers' convenience using a genetic algorithm to trade-off the total cost of road-intermodal and GHG emissions.
- To compare the R1, R2, and R3 routes in road transportation with the intermodal, and to recommend the most cost-effective and environmentally friendly route.

Problem Statement

This section describes the design and definition of the bi-objective problem, as well as the formulation of the MILP model to investigate the cost and GHG emissions of road and road-rail freight transportation.

Problem Definition

This section defines our bi-objective MILP problem, with the main objective of minimizing the total cost of road and road-rail freight transportation in Pakistan from dry ports to seaports. Our second objective is to minimize GHG emissions in the same context. Logistics operations in Pakistan begin at critical dry ports in various cities, and unit containers are used for a single type of product. The consignments are transported by a single type of truck with capacity to carry a 20 ft container between the origin and destination pairs. In road freight transportation, the consignment is loaded at one of various dry ports throughout Pakistan and offloaded at the seaport in Karachi (Sindh). Figure 1 depicts the geographic locations of the dry ports and seaport. However, in intermodal freight distribution, containers are taken from one of the various dry ports to the nearest railway station by truck, as the "first-mile" delivery. After loading on to a freight train, the consignment is transported by rail from the origin to the destination. In the "last-mile" delivery, the cargo is unloaded at Karachi Railway Station and transported by truck to the seaport of Karachi.

The case studies are created using real-world data from one of Pakistan's largest logistics service providers. In practice, logistics provision begins at various dry ports distributed across Pakistan's provinces. Pakistan has a total of 12 active dry ports. Trucks travel to their destinations via different road distribution routes (shortest and longest) from each city.

Modeling. The bi-objective MILP model is presented in this section to formulate the green road and intermodal

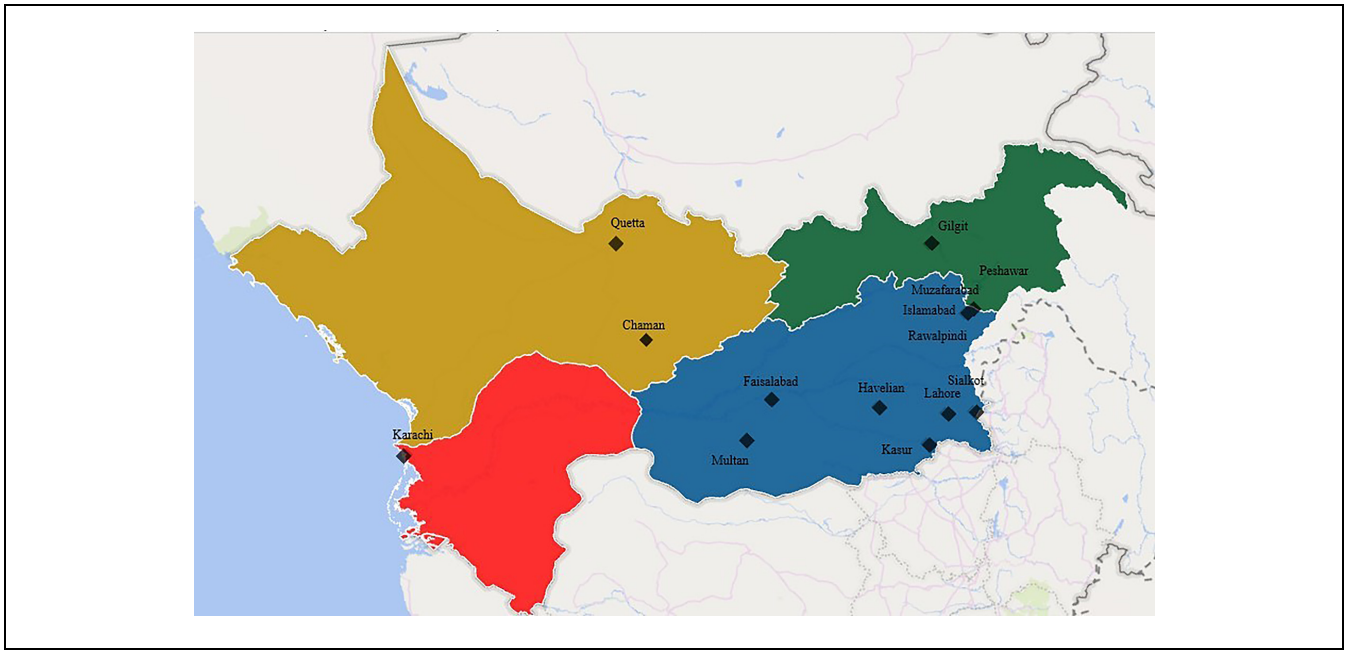


Figure 1. Geography of dry ports and seaport in Pakistan.

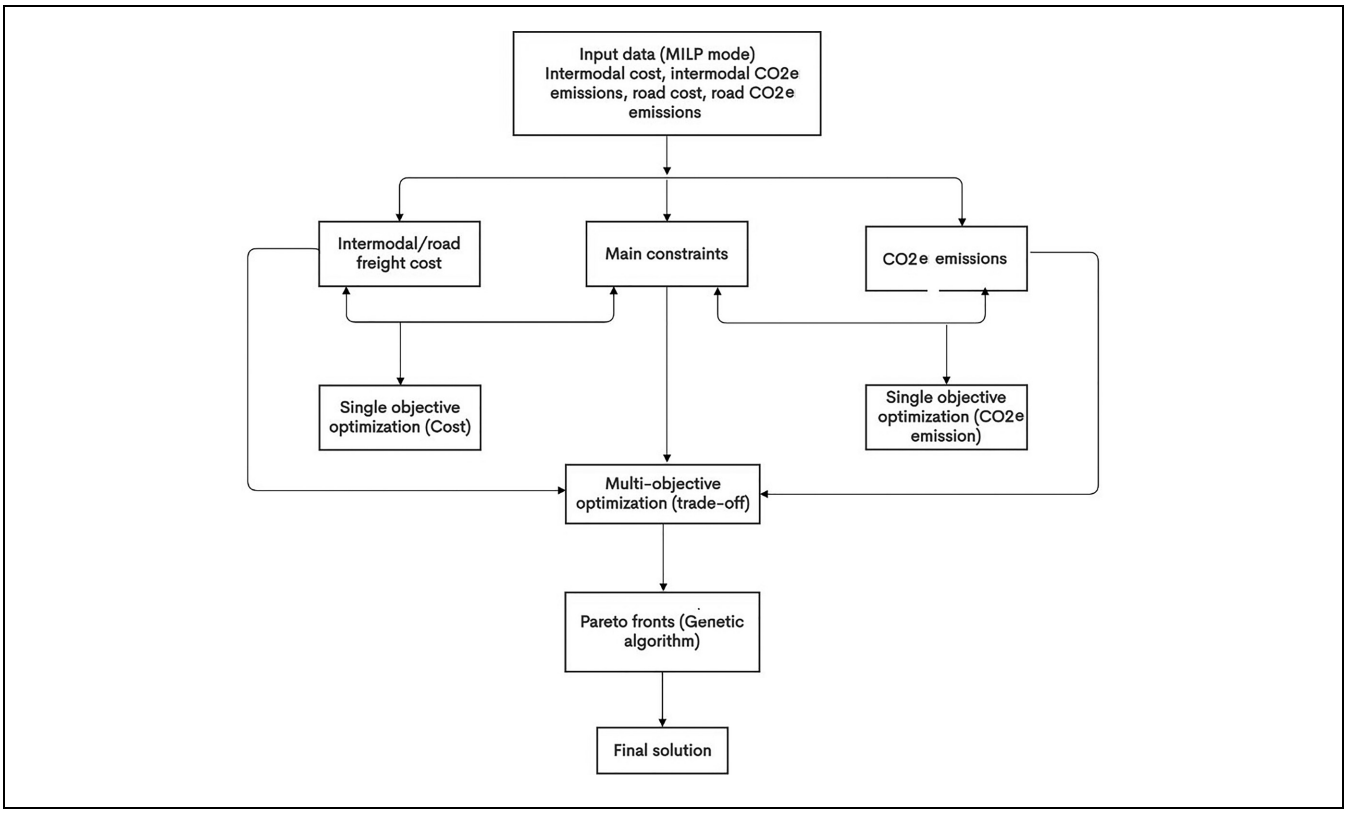


Figure 2. Flow chart of research methodology.

distribution network design problem. Our model is based on the calculations presented by (Demir et al. (*46*)), and it presents environmental and economic objectives that have been optimized using a genetic algorithm. Figure 2 depicts a flowchart of the research methodology.

The model's goal is to generate the best plan possible for the orders $o \in O$ outlined by the demand d^p between the origin node i to destination node j. Furthermore, $\gamma^p (i,j) = \{(o \in O) \mid i \in TN^+ \text{ and } j \in TN^-\}$ shows the set of orders from the source i to destination j. TN^+ shows the set of start and TN^- the set of end terminals of transportation orders. The transportation network consists of nodes $i, j \in NT$ for transportation locations and services $s \in S$. Notations for the sets and parameters are as follows:

TN = Set of all locations of transshipment
TN^+ = Set of start stations for transporting orders
TN^- = Set of end stations for transporting orders
O = Set of transportation of orders
S = Set of services provided by transportation
c_j = Cost of transshipment per container at station j
Rc^s = Road cost of transportation per container of a service s
c^s = Cost of transportation per container per service s
Rc_e = Cost of GHG emissions per kilogram
dm^o = The demand of container for order o
em_j = Emissions per transshipment of container in kilograms at station j
em^s = Emissions in the transportation of container in kilograms for service s
rem^s = Emissions in kilograms in road transportation of container on service s
L = Large number.

Notations for decision variables are as follows:

n_j = Total containers transshipped at station j
y^s, y^{so} = A binary variable = 1 for order o, if service s is used
x^{so} = Use of service s for number of containers of order o
z^{qro} = Transshipment of the containers between the proceeding and succeeding services q and r.

$$minimize\ f_1(x) = \sum_{o\in O}\sum_{s\in S} x^{so} Rc^s \tag{1}$$

$$minimize\ f_2(x) = \sum_{o\in O}\sum_{s\in S} x^{so} c^s + \sum_{j\in TN} n_j c_j \tag{2}$$

$$minimize\ f_3\ (x) = \left(\sum_{o\in O}\sum_{s\in S} x^{so} rem^s\right) \tag{3}$$

$$minimize\ f_4\ (x) = \left(\sum_{o\in O}\sum_{s\in S} x^{so} em^s + \sum_{j\in TN} n_j em_j\right) \tag{4}$$

Subject to:

$$\sum_{s\in\delta(s\in S|n=i)} x^{so} = dm^{o}\ \forall n\ \in TN|n=i), o\in O \quad (5)$$

$$\sum_{s\in\delta(s\in S|n=j)} x^{so} = dm^{o}\ \forall n\ \in TN|n=j), o\in O \quad (6)$$

$$\sum_{s\in\delta(s\in S|n=i)} x^{so} - \sum_{s\in\delta(s\in S|n=j)} x^{so} = 0\forall n\ \in TN|n=i,j),\ o\in O \quad (7)$$

$$x^{so} \leqslant y^{so}\ \forall s\in\delta(s\in S), o\in\gamma(o\in O) \quad (8)$$

$$x^{so} \leqslant y^{so}\ \forall s\in\delta(s\in S),\ o\in\gamma(o\in O) \quad (9)$$

$$y^{s} \leqslant \sum_{p\in\gamma(o\in O)} x^{so}\ \forall s\in\delta(s\in S) \quad (10)$$

$$y^{s}, y^{so}\in\{0,\ 1\}\ \forall s\in S,\ q\in S,\ r\in S,\ o\in O \quad (11)$$

$$x^{so},\ z^{qro}\geqslant 0\ q\in S,\ r\in S,\ o\in O \quad (12)$$

Equation 1 shows the cost of road transportation of goods from dry port to seaport. Equation 2 shows the total GHG emissions in road freight distribution. Equations 3 and 4 depict the cost and GHG emissions, respectively, of intermodal freight transportation. Equations 5 and 6 focus on the containers' origin and destination nodes and movement. Equation 7 manages the transshipment from dry ports to the seaport. Equations 8 to 10 ensure that only one service is acceptable to process any number of containers when it is selected. Equations 11 and 12 show the domain of the decision variables of the problem.

Solution Methodology. In this study, we used a genetic algorithm developed by Holland (*47*) to find the solutions of the multi-objective MILP problem. The problem is focused on four objective functions related to reducing the total cost of logistics from Pakistan's dry ports to Karachi seaport via road and intermodal freight transportation. We also looked at GHG emissions from road and rail freight transportation. The cost of logistics is most important in the transportation sector. As a result, we prioritized the cost of road and road-rail freight. In both modes of transportation, we used Pareto-optimal solutions to trade-off costs and GHG emissions.

Genetic algorithms (GA) are widely used nowadays to obtain high-quality solutions for problem optimization by relying on operators such as selection, crossover, and mutation. The significance of using GAs cannot be overstated given their ability to be implemented as a universal optimizer that can be used to optimize any optimization problem, as well as their ease of implementation, proper stability between exploration and exploitation by correctly setting the parameters, and logical reasoning behind the use of operators.

GA applies the concept of the fittest cross-breeding population's existence to develop a robust search strategy. For this purpose, first, the solutions with the finite population by taking the specified problem are maintained. Second, new populations are created from existing populations by prioritizing the solutions and ranking them according to their fitness values and cross-breeding of fittest again to obtain the most optimized new offspring. GA only uses the fitness value for its operations without any other knowledge. The solutions are produced as generations, and weaker ones are eliminated without having offspring. However, the more robust solutions remain and combine parents' features to develop new families, and the cycle continues. In optimizing the GA problem, the fitness function is needed to select the best candidates for the population's solution and eliminate the non-supporting specimens. After defining the genetic representation and fitness function, the GA initializes the population of solutions randomly. It then improves the solutions repetitively by applying operators like mutations, crossover, and selection. In this research, the developed mathematical model with GA application is written as M-files using MATLAB software. Steps in the implementation of the GA are as follows:

1. Evaluation of each individual in the targeted population
2. Selection of the best and fittest parents out of the given population
3. Performing the crossover by combining the individuals from parents again to produce their children as a new generation
4. Mutation of the newly created generation
5. If no termination, repeat step 2 until the operation's termination and return to the best-found individual in the present population.

GA finds a set of solutions within a range with a good spread. The fitness function is constructed in multi-

Bi-objective genetic algorithm pseudocode

```
Begin
    t = 0
Initialize the population for chromosomes C(gg);
    Initialize population's evaluation by calculating its fitness
    C(gg);
While criteria of termination not met
gg: = gg + 1;
    Select C(gg + 1) from C(gg);
Crossover C(gg + 1);
Mutate C(gg + 1);
    Evaluate C(gg + 1);
 End While
Output results to external archive
 End
```

objective GA (MOGA). The fitness function optimizes two objective functions simultaneously using Matlab's optimization toolbox. Using Matlab's toolbox, the single objective can be optimized. All Pareto solutions are optimal. Inputs are x_1, x_2, x_3, $x_4 \ldots x_n$ of variable "x." There is only one input value x in the fitness function, and it has as many elements as the problem variables. The fitness function calculates and returns each objective function's value as a single vector "y."

The set of variables and fitness function are required to implement the MOGA optimization issue using the Matlab toolbox. The MOGA's first two output arguments are x and fval, which show the Pareto front's points and values. The output exitflag's third argument stops MOGA. The fourth argument output shows the algorithm's performance. MOGA returns the fifth argument, population. This argument contains the MOGA population. When the solver ends, the score contains the function values of all the objectives.

GA encodes decision variables or parameters, generating a population of initial solutions. GA affects the whole population at once. A random generator is used to create the desired number of beginning individuals or populations. This is the most popular approach, where each vector is encoded as a binary string and then concatenated to form chromosomes. We can use a bitstring or a double vector to choose chromosomes in a population. If the population chooses the bitstring option, the solver automatically decodes the data.

After the optimization, iterations begin, and the solver toolbox displays the functions and decision variables. Options include population type (double vector or bitstring), crossover (constraint dependent, single, double, or dispersed), mutation (uniform or gaussian), and additional optimizations.

Computational Experiments

This section discusses the primary data applied in the calculation and analysis, and presents a general overview of the various real-life cases of road and road-rail transportation moving goods from origin to destination, the application of the cases in the study, and some assumptions applied in the operation and design of the road and intermodal distribution networks.

This research applies real-life data from one of the largest logistics services providers in Pakistan. The data include the locations of critical dry ports, hubs of unloading and loading operations, links between all locations, distances between the locations and links, different types of vehicles, loading and off-loading, capacities of vehicles, historical data of supply and demand, and some limits in road and rail operations. Moreover, the existing movements from 12 dry ports to a seaport by road and rail are considered for this research. Demand at the seaport is taken as constant.

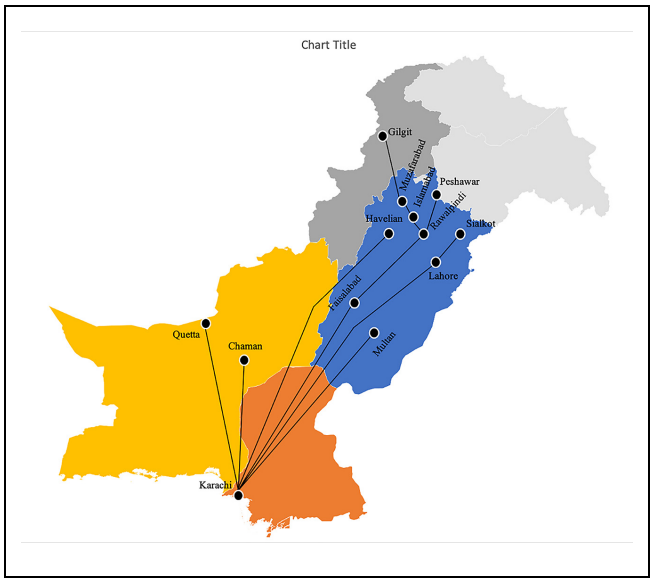


Figure 3. Route of road freight transportation (R1) from dry ports to seaport, Karachi.

Results

This section looks at the MILP model through the lens of various scenarios. The number of nodes for short (R1), medium (R2), and long (R3) road freight distribution routes is the key difference in all three cases. Routes R3, R2, and R1 have five, four, and two nodes, respectively. Figure 3 shows the direct movement of cargo from dry ports to the seaport with two nodes (black). Figure 4 depicts the indirect movement of cargo with totals of four and five nodes (black and red). R3 is the longest route for moving cargo from dry ports to the seaport, R2 is an average distance, and R1 is the shortest route for road freight transportation. In the intermodal freight distribution, there are two transshipment points. Pareto front solutions were obtained by implementing real-life data using a GA. The model is run on a computer with a 2.50 GHz dual-core processor and 4.00 GB of RAM installed. The necessary parameters of the MOGA are shown in Table 2.

Case #1 (Road Freight Distribution)

The graphical representation of the case is illustrated in Figures 3 and 4. For road freight transportation, three routes were calculated from each of 12 dry ports to the seaport. The shortest route (R1) from all dry ports is 1,269 km long on average. In Pakistan, the average cost of road freight distribution for the delivery of 16 tonnes

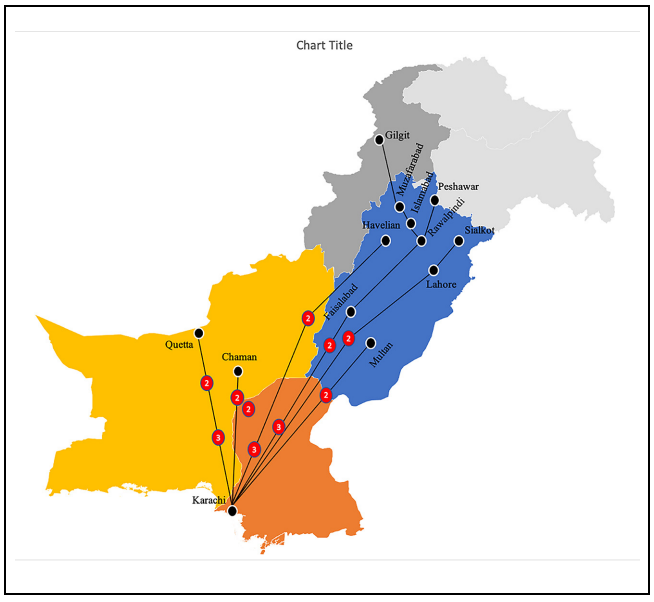


Figure 4. Route of road freight transportation (R2) from dry ports to seaport Karachi.

Table 2. Values of Parameters of Genetic Algorithm

Genetic algorithm parameter	Value
Population size	100
Population type	Double vector
Number of iterations	167
Crossover rate	0.76
Creation function	Constraint dependent
Mutation	Uniform
Crossover	Constraint dependent
Selection function	Tournament
Crossover fraction	0.8
Migration direction	Forward
Number of variables	4

of material in a 20 ft container is 3.3 Pakistani rupee Pkr/tonne/km. Therefore, the average cost of transportation to deliver 16 tonnes of material between the dry ports and Karachi seaport using R1 is 66,990 Pkr. The pollution factor developed by Tardif (*48*) to benchmark fuel efficiency in Canada's trucking industry, as well as Canada's national CO_2 inventory report from 1990 to 2018, was used to calculate road emissions (*31*). A weight of 16 tonnes is used in the CO_2e calculation to generate an emission factor of 63.8 g CO_2e/tonne-km. For rail transport, the rail emission factor is calculated using the average rail consumption (liter diesel/revenue tonne-kilometer) in 2017, as reported by the Railway Association of Canada's Locomotive Emissions Monitoring Program, which provides data on average diesel consumption per tonne-kilometer in rail transportation (*31*). We used these conversion factors to calculate CO_2e emissions for Pakistan's road and rail freight transportation. The overall CO_2e emission factor is 14.0 g CO_2e/tonne-km. Therefore, the average CO_2e emissions of a 16 tonne delivery from the 12 dry ports to the seaport of Karachi are 1.27 tonnes for R1.

The average length of R2 from the 12 dry ports to the seaport of Karachi is 1,460 km. In the delivery of 16 tonnes of material using R2, the average cost and CO_2e emissions are 77,109 Pkr and 1.47 tonnes, respectively. On the same basis, R3 (the longest route) has an average length of 1,499 km, cost of 79,168 Pkr, and CO_2e emissions of 1.5 tonnes. Key data for the calculations are available in Tables 3 to 5.

Case #2 (Intermodal Freight Distribution)

The graphical illustration of intermodal freight distribution is shown in Figure 5. In this case, there are two transshipment points. To make an apples-to-apples comparison with road freight transportation, intermodal freight transportation was calculated also using the average distance, cost, and CO_2e emissions. The distances from all the respective dry ports to the Karachi seaport are given in Tables 6 and 7. Because of Pakistan's limited rail infrastructure, only one route from each dry port to Karachi was studied. This case study comprises a total of 12 dry ports, each of which regularly transports goods to Karachi Railway Station. In the "first-mile" delivery, freight forwarders transport cargo containers by road from each of the dry ports to a nearby railway station. The containers are unloaded from trucks at the railway station and loaded onto a train bound for Karachi. The containers are unloaded at Karachi Railway Station and reloaded onto trucks for the "last mile", the final leg of the journey. Finally, the containers are unloaded at the seaport of Karachi. In the movement of containers from the dry ports to the seaport, the average distance covered by trucks as the "first-mile" delivery is 13 km (see Table 6).

The average distance from the dry port railway station to Karachi Railway Station is 1,302 km and the "last-mile" delivery distance, from Karachi Railway Station to the seaport, is 7 km (Table 7). The average cost of delivering 16 tonnes of material is 3.3 Pkr/tonne for road freight transportation and 0.45 Pkr/tonne by rail. The railway freight cost data were obtained from the official website of Pakistan Railways https://www.pakrail.gov.pk/. The average cost and CO_2e emissions for the "first-mile" delivery are 872 Pkr and 0.013 tonnes. For the rail leg of the journey, the average cost and CO_2e emission are 11,719 Pkr and 0.29 tonnes. For the "last-mile" delivery, the average cost and CO_2e emissions are 369 Pkr and 0.007 tonnes. Therefore, the total cost of intermodal freight distribution is 12,961 Pkr and 0.31 tonnes

Table 3. Road Freight Transportation (R1)

No.	Dry port	Node	Distance (km)	Cost (Pkr)	GHG emissions (g CO_2e)
1	Islamabad	1	1,409	74,395	1,420,272
2	Chaman Border Terminal	2	810	42,768	816,480
3	Quetta Dry Port	3	587	30,994	591,696
4	Muzaffarabad Dry Port	5	1,582	83,530	1,594,656
5	Gilgit Dry Port	6	1,883	99,422	1,898,064
6	Havelian Dry Port	8	1,486	78,461	1,497,888
7	Peshawar Dry Port	9	1,553	81,998	1,565,424
8	Faisalabad Dry Port	11	1,115	58,872	1,123,920
9	Lahore Dry Port	12	1,210	63,888	1,219,680
10	Multan Dry Port	13	886	46,781	893,088
11	Rawalpindi Dry Port	14	1,393	73,550	1,404,144
12	Sialkot Dry Port	15	1,311	69,221	1,321,488
Average			1,269	66,990	1,278,900 (1.27 tonnes)

Note: GHG = greenhouse gases.

Table 4. Road Freight Transportation (R2)

No.	Dry port	Nodes	Distance (km)	Cost (Pkr)	GHG emissions (g CO_2e)
1	Islamabad	17, 18	1,435	75,768	1,446,480
2	Chaman Border Terminal	22, 23	977	51,586	984,816
3	Quetta Dry Port	24, 25	847	44,722	853,776
4	Muzaffarabad Dry Port	31, 32	1,606	84,797	1,618,848
5	Gilgit Dry Port	36, 37	2,069	109,243	2,085,552
6	Havelian Dry Port	43, 44	1,510	79,728	1,522,080
7	Peshawar Dry Port	48, 49	1,407	74,290	1,418,256
9	Lahore Dry Port	52, 53	1,610	85,008	1,622,880
11	Rawalpindi Dry Port	54, 55	1,440	76,032	1,451,520
12	Sialkot Dry Port	59, 60	1,703	89,918	1,716,624
Average			1,460	77,109	1,472,083 (1.47 tonnes)

Note: GHG = greenhouse gases.

Table 5. Road Freight Transportation (R3)

No.	Dry port	Nodes	Distance (km)	Cost (Pkr)	GHGs emissions (g CO_2e)
1	Islamabad	19, 20, 21	1,597	84,322	1,609,776
2	Quetta Dry Port	26, 27, 28	857	45,250	863,856
3	Muzaffarabad Dry Port	33, 34, 35	1,769	93,403	1,783,152
4	Havelian Dry Port	45, 46, 47	1,672	88,282	1,685,376
5	Rawalpindi Dry Port	56, 57, 58	1,602	84,586	1,614,816
Average			1,499	79,168	1,511,395 (1.5 tonnes)

Note: GHG = greenhouse gases.

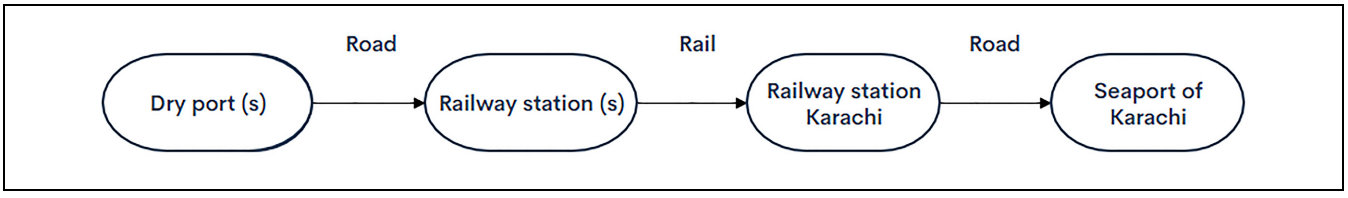


Figure 5. Graphical representation of intermodal freight transportation.

Table 6. Intermodal Freight Transportation (A)

No.	Dry port	"First mile" distance (km)	CO_2e	Cost (Pkr)
1	Islamabad	2.6	2,620.8	171.6
2	Chaman	18	18,144	1,188
3	Quetta	5	5,040	330
4	Muzafarabad	19	19,152	1,254
5	Gilgit	22	22,176	1,452
6	Havelian	17	17,136	1,122
7	Peshawar	2	2,016	132
8	Faisalabad	13	13,104	858
9	lahore	14	14,112	924
10	Multan	21	21,168	1,386
11	Rawalpindi	7	7,056	462
12	Sialkot	18	18,144	1,188
Average		13	13,322	872

of CO_2e emissions to deliver 16 tonnes of material from a dry port to the seaport of Karachi.

The difference in cost of road transportation versus intermodal freight distribution is: 66,990 − 12,961 = 54,029 Pkr (R1), 77,109 − 12,961 = 64,148 Pkr (R2), and 79,168 − 12,961 = 66,207 Pkr (R3). The total savings of intermodal freight distribution over road transportation therefore is 80.6% (R1), 83.1% (R2), or 83.6% (R3). The difference in the CO_2e emissions of road transportation versus intermodal freight distribution is: 1.27 − 0.31 = 0.96 tonnes (R1), 1.47 − 0.31 = 1.16 tonnes (R2), and 1.5 − 0.31 = 1.19 tonnes (R3). Therefore, the total savings of CO_2e emissions from intermodal freight distribution over road freight transportation are: 75.5% (R1), 79% (R2), and 79.3% (R3).

The average cost and CO_2e emissions of all the road routes (R1, R2, and R3) are 74,422 Pkr and 1.41 tonnes, compared with intermodal freight distribution at 74,422 − 12,961 = 61,461 Pkr and 1.41 − 0.31 = 1.1 tonnes. Therefore, intermodal freight distribution is 82.6% more economical and 78% greener than road transportation.

Figures 6 and 7 show the GA results (left to right) as follows: i. the average distance of all the individuals in every generation; ii. in each generation, the cross-sectional results of individuals; iii. the values of objective function generated at the end of the algorithm operation; iv. the total progeny ("children") generated by every individual; v. change in the values of the objective functions; vi. in the last generation, the distance of each individual; vii. in the final population, the distribution of individual fitness; viii. at each run, the average speed of the algorithm. Tables 8 and 9 show the computational experiment results of GHG emissions and cost in the road and intermodal transportation networks.

GAs, unlike chaotic algorithms, are stochastic. Individual and population representation, fitness function, and genetic operators all contribute to the concept of complexity. As to computing time, Table 10 displays the performance of the MOGA for both objective functions. This study, however, makes no comparisons with other multi-objective optimization algorithms such as multi-objective particle swarm optimization or multi-objective bat algorithm. Both the objective functions for cost and GHG emissions minimization are studied in the generation of Pareto-optimal solutions in this performance analysis. The time efficiency shows that MOGA running time is between 40 and 46 s for cost function and between 37 and 42 s for minimization of GHG emissions with approximately 200 iterations for each run.

The computational time experiment is conducted on a computer with a 2.50 GHz dual-core processor and 4.00 GB of RAM. The MOGA is run 25 times and the computing time is shown in Figure 8.

Conclusion

There are many studies based on traditional methods of reducing transportation costs from suppliers to

Table 7. Intermodal Freight Transportation (B)

No.	Dry port	Second stage distance (km)	CO_2e	Cost (Pkr)	"Last mile" distance (km)	CO_2e	Cost (Pkr)
1	Islamabad	1,521	340,704	13,689	7	7056	369.6
2	Chaman	810	181,440	7,290	7	7056	369.6
3	Quetta	693	155,232	6,237	7	7056	369.6
4	Muzafarabad	1,583	354,592	14,247	7	7056	369.6
5	Gilgit	1,886	422,464	16,974	7	7056	369.6
6	Havelian	1,485	332,640	13,365	7	7056	369.6
7	Peshawar	1,554	348,096	13,986	7	7056	369.6
8	Faisalabad	1,106	247,744	9,954	7	7056	369.6
9	Lahore	1,246	279,104	11,214	7	7056	369.6
10	Multan	881	197,344	7,929	7	7056	369.6
11	Rawalpindi	1,548	346,752	13,932	7	7056	369.6
12	Sialkot	1,312	293,888	11,808	7	7056	369.6
Average		1,302	291,667	11,719	7	7056	369.6

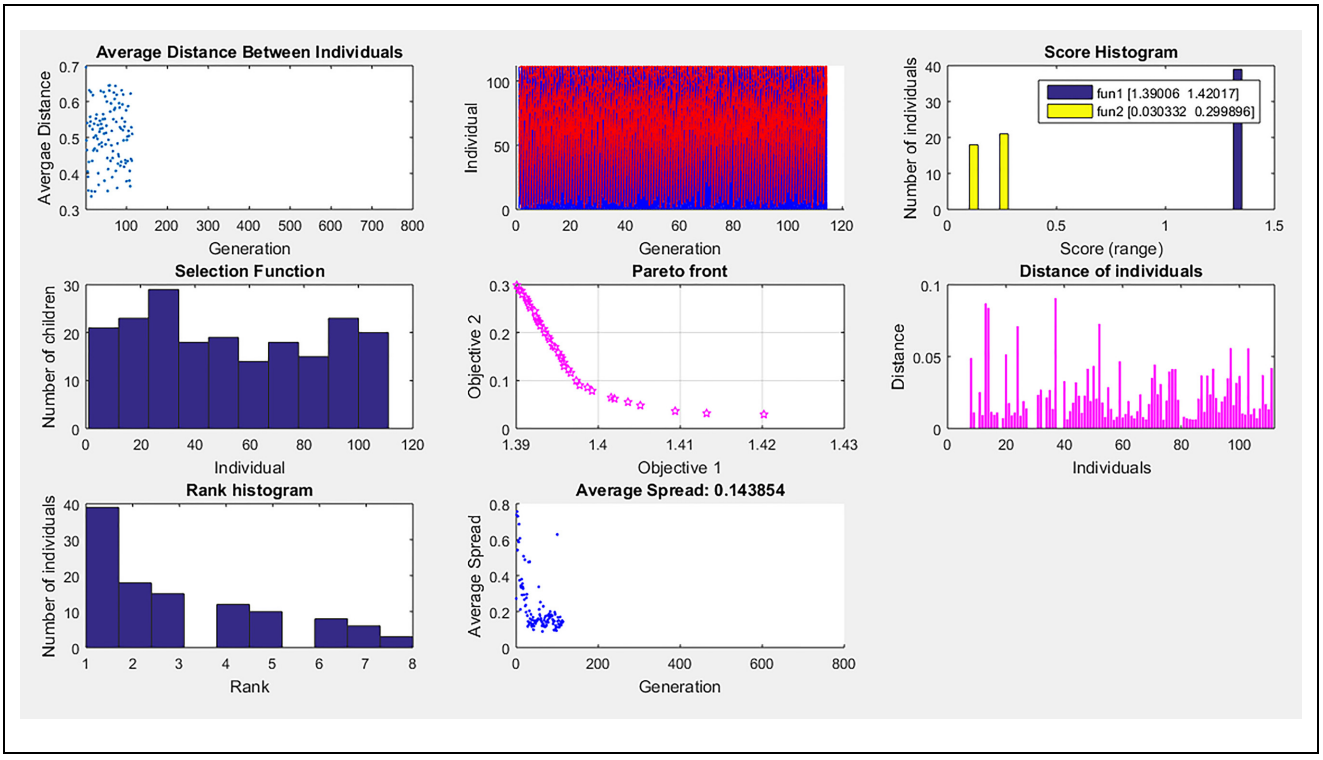


Figure 6. Genetic algorithm results for greenhouse gas emissions from road and intermodal freight transportation.

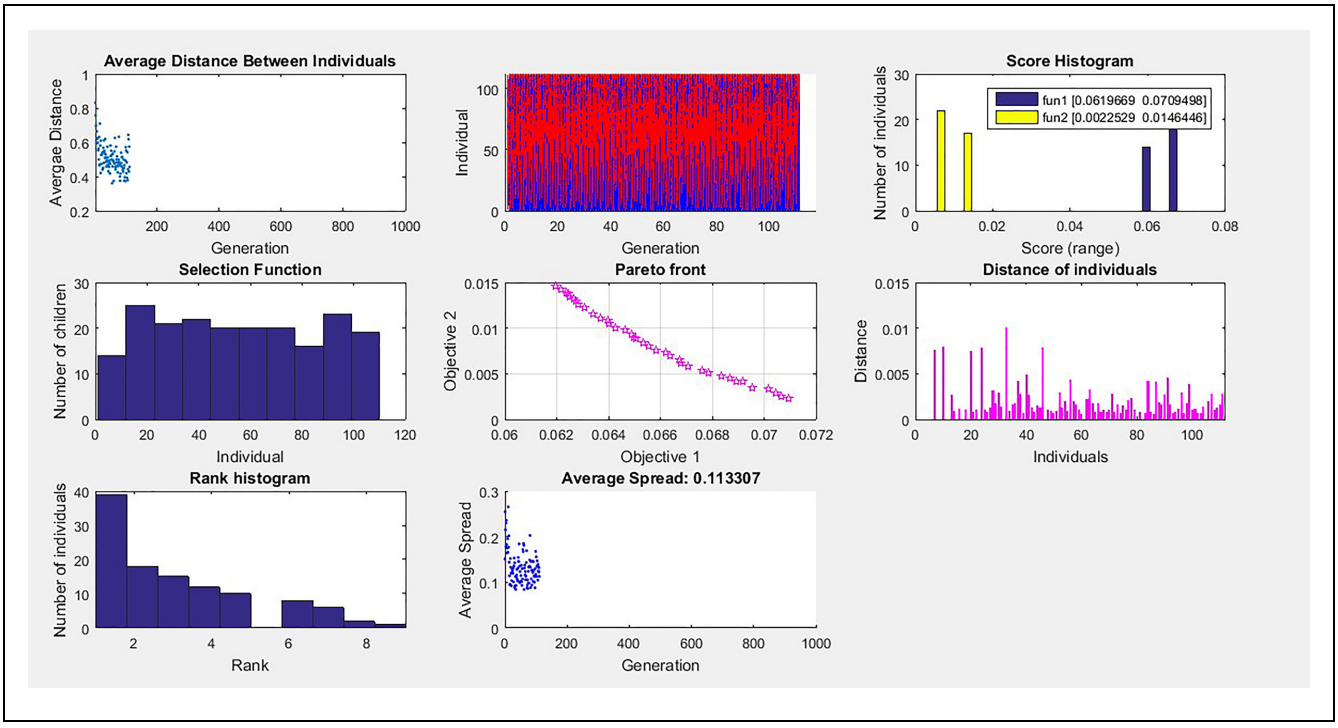


Figure 7. Genetic algorithm results for costs of road and intermodal freight transportation.

customers by road, rail, sea, and air. However, studies that take green logistics into account in addition to reducing transportation costs are limited. Our study improves the split of the modes, minimizing both the costs and CO_2e emissions of transportation between dry ports and seaports. The focus on green logistics and costs

Table 8. Results of Computational Experiments (Greenhouse Gas Emissions in Road and Intermodal Transportation)

Initial population (double vector)				Initial score		Initial population (bitstring)			
x_1	x_2	x_3	x_4	f_1	f_2	x_1	x_2	x_3	x_4
0.987917	0.007982	0.004646	0.005802	1.400559	0.306804	0	0	0	0
0.005331	0.011088	0.009582	0.024214	1.410988	0.002641	0	0	0	0
0.997067	0.009893	0.004942	0.009722	1.400595	0.309781	1	1	0	0
0.005331	0.011088	0.009582	0.024214	1.410988	0.002641	0	0	0	1
0.005331	0.011073	0.008894	0.023853	1.410956	0.002623	1	0	0	0
0.997067	0.072393	0.008848	0.009722	1.401981	0.312985	0	0	0	1
0.005385	0.011059	0.007888	0.022182	1.410887	0.002602	0	0	0	1
0.991952	0.009649	0.004814	0.00631	1.400568	0.308147	0	0	0	0
0.847207	0.048145	0.018337	0.038419	1.403901	0.265792	1	0	0	1
0.00534	0.011077	0.008576	0.024113	1.410951	0.002622	1	1	0	0
0.862917	0.101732	0.051521	0.021427	1.405637	0.273836	0	0	1	0
0.04932	0.01058	0.007311	0.013703	1.410248	0.016101	1	0	0	0
0.346082	0.00956	0.008858	0.019072	1.407422	0.108131	0	1	0	0
0.477451	0.034047	0.011252	0.015282	1.406606	0.15009	1	0	0	0
0.97488	0.068518	0.008871	0.010729	1.402147	0.305923	1	0	0	0
0.634046	0.010603	0.006076	0.010756	1.404299	0.197314	0	1	0	0
0.990802	0.009287	0.004798	0.005997	1.400566	0.307769	1	0	0	0
0.677591	0.008205	0.008502	0.015703	1.404	0.21079	1	1	0	0
0.353302	0.011229	0.008726	0.02977	1.407592	0.110557	1	0	0	1
0.117702	0.050675	0.008115	0.017258	1.410466	0.039356	1	0	0	0
0.53391	0.010594	0.010019	0.023825	1.4057	0.16648	0	0	0	1
0.617397	0.060719	0.010307	0.017618	1.405754	0.194811	0	0	0	1
0.005345	0.011073	0.008202	0.023782	1.410931	0.002612	0	0	0	1
0.582262	0.011997	0.007988	0.011963	1.404936	0.18138	0	0	0	1
0.005375	0.011073	0.00815	0.022868	1.41091	0.002612	0	0	0	0
0.459693	0.010788	0.008722	0.017841	1.406281	0.143397	0	0	0	0
0.478271	0.020253	0.012124	0.022469	1.406496	0.149744	0	1	0	0
0.403734	0.010775	0.010189	0.017264	1.40688	0.126073	0	0	0	0
0.675821	0.011333	0.007414	0.010463	1.403937	0.210324	0	0	0	0
0.309413	0.044607	0.006649	0.052458	1.40908	0.098806	0	0	0	1
0.992966	0.009846	0.004933	0.007409	1.400588	0.308484	1	0	0	0
0.135361	0.010958	0.008289	0.074909	1.410654	0.043425	1	0	0	0
0.936855	0.008775	0.006916	0.011254	1.401274	0.291115	0	0	0	0
0.454925	0.011323	0.009328	0.025257	1.406509	0.142032	0	1	0	0
0.733377	0.010247	0.006536	0.062553	1.404351	0.228615	0	0	0	1
0.987917	0.007982	0.004646	0.005802	1.400559	0.306804	1	0	0	1
0.286994	0.010567	0.008155	0.025276	1.408132	0.089912	0	0	1	0
0.005385	0.011059	0.007888	0.022182	1.410887	0.002602	1	0	0	1
0.105176	0.011051	0.007788	0.022976	1.409901	0.033543	0	0	0	1
0.334488	0.011276	0.007787	0.011675	1.407387	0.104527	0	1	0	0
0.791156	0.03114	0.009849	0.016764	1.403391	0.24718	0	0	1	0
0.384294	0.043312	0.010052	0.02435	1.407862	0.121741	1	1	0	0
0.232027	0.017312	0.007254	0.013361	1.408547	0.073073	0	0	0	1
0.85541	0.012633	0.007823	0.019284	1.402358	0.266158	0	0	1	0
0.439367	0.010982	0.008436	0.013744	1.406396	0.137059	1	1	0	0
0.163558	0.021763	0.005339	0.009958	1.409186	0.051998	1	0	0	1
0.365425	0.013428	0.009622	0.062795	1.408207	0.114773	0	1	0	0
0.150011	0.009599	0.00872	0.020917	1.409415	0.047367	0	0	0	1
0.350738	0.011097	0.008296	0.023026	1.407465	0.10968	0	0	0	0
0.651747	0.084844	0.00846	0.269367	1.410863	0.209147	1	0	0	0

Note: Average spread = 0.091; average distance = 0.0033, total generations = 233; Pareto fraction = 0.35; crossover fraction = 0.80; max stall generations = 100.

can help in planning transportation as this topic is gaining much attention.

The purpose of this study is to assist decision makers, practitioners, and industrialists in increasing flexibility, productivity, safety, and forecasting. The study's ultimate objective is to reduce the cost of road and intermodal freight transportation between Pakistan's seaports and dry ports. A secondary goal is to provide environmentally

Table 9. Results of Computational Experiments (Cost in Road and Intermodal Transportation)

Initial population (double vector)				Initial score		Initial population (bitstring)			
x_1	x_2	x_3	x_4	f_1	f_2	x_1	x_2	x_3	x_4
0.197939	0.00551	0.035005	0.065404	0.010437	0.015441	0	0	0	0
0.639489	0.007137	0.035005	0.065404	0.006054	0.048123	0	1	0	0
0.639489	0.00551	0.035005	0.065404	0.006021	0.048115	1	0	1	0
0.639489	0.009552	0.035005	0.065404	0.006102	0.048135	1	0	0	1
0.778973	0.00551	0.035005	0.065404	0.004626	0.058437	0	0	0	0
0.023577	0.00551	0.035005	0.065404	0.01218	0.002538	0	1	0	0
0.023577	0.009552	0.035005	0.065404	0.012261	0.002558	0	0	1	0
0.023577	0.007137	0.035005	0.065404	0.012213	0.002546	0	0	1	0
0.912356	0.00551	0.035005	0.065404	0.003292	0.068307	1	0	0	0
0.325311	0.00551	0.035005	0.065404	0.009163	0.024866	0	0	0	0
0.778973	0.009552	0.035005	0.065404	0.004707	0.058457	0	0	0	1
0.325311	0.007137	0.035005	0.065404	0.009195	0.024874	0	0	0	0
0.197939	0.007137	0.035005	0.065404	0.010469	0.015449	1	0	0	1
0.778973	0.007137	0.035005	0.065404	0.004659	0.058445	0	0	0	0
0.841213	0.00551	0.035005	0.065404	0.004004	0.063043	1	0	0	1
0.325311	0.009552	0.035005	0.065404	0.009244	0.024886	0	0	0	1
0.912356	0.00551	0.035005	0.080894	0.003323	0.068323	1	0	0	0
0.912356	0.007137	0.035005	0.080894	0.003356	0.068331	1	0	0	1
0.477956	0.007137	0.035005	0.065404	0.007669	0.03617	1	0	0	1
0.477956	0.00551	0.035005	0.065404	0.007636	0.036162	1	0	0	0
0.477956	0.007137	0.035005	0.080894	0.0077	0.036185	1	0	0	0
0.477956	0.007137	0.035005	0.080894	0.0077	0.036185	1	0	0	1
0.197939	0.007137	0.035005	0.080894	0.0105	0.015464	0	0	0	0
0.841213	0.009552	0.035005	0.065404	0.004085	0.063063	1	0	0	0
0.08825	0.00551	0.035005	0.065404	0.011534	0.007324	1	0	0	0
0.334852	0.007137	0.035005	0.080894	0.009131	0.025596	1	0	1	0
0.197939	0.009552	0.035005	0.065404	0.010517	0.015461	0	0	0	0
0.668017	0.007137	0.035005	0.080894	0.005799	0.05025	1	0	0	0
0.197939	0.00551	0.035005	0.080894	0.010468	0.015456	0	0	0	0
0.023577	0.00551	0.035005	0.080894	0.012211	0.002553	0	0	0	0
0.08825	0.007137	0.035005	0.065404	0.011566	0.007332	0	0	0	1
0.023577	0.009552	0.035005	0.065404	0.012261	0.002558	0	0	1	0
0.334852	0.00551	0.035005	0.065404	0.009068	0.025572	1	0	0	1
0.08825	0.00551	0.035005	0.080894	0.011565	0.007339	0	1	0	0
0.841213	0.00551	0.035005	0.080894	0.004035	0.063058	0	0	0	1
0.912356	0.00551	0.035005	0.080894	0.003323	0.068323	1	0	0	1
0.08825	0.00551	0.035005	0.065404	0.011534	0.007324	0	0	1	0
0.023577	0.007137	0.035005	0.065404	0.012213	0.002546	1	0	1	0
0.023577	0.007137	0.035005	0.080894	0.012244	0.002561	1	0	0	1
0.334852	0.00551	0.035005	0.080894	0.009098	0.025588	0	0	0	1
0.668017	0.00551	0.035005	0.065404	0.005736	0.050226	1	0	1	0
0.841213	0.00551	0.035005	0.065404	0.004004	0.063043	0	1	0	0
0.912356	0.00551	0.035005	0.065404	0.003292	0.068307	0	0	0	1
0.790766	0.00551	0.035005	0.065404	0.004508	0.05931	0	1	0	0
0.477956	0.00551	0.035005	0.080894	0.007667	0.036177	0	0	0	1
0.912356	0.007137	0.035005	0.080894	0.003356	0.068331	1	0	0	1
0.668017	0.00551	0.035005	0.080894	0.005767	0.050242	1	0	1	0
0.477956	0.00551	0.035005	0.065404	0.007636	0.036162	0	0	1	0
0.023577	0.00551	0.035005	0.065404	0.01218	0.002538	0	0	0	0
0.197939	0.00551	0.035005	0.065404	0.010437	0.015441	1	0	0	0

Note: Average spread = 0.164; average distance = 0.0028; total generations = 233, Pareto fraction = 0.35, crossover fraction = 0.80; max stall generations = 110.

friendly transportation and logistics by reducing CO_2 emissions. We employed MILP to formulate the cost and CO_2e emissions problem. We compared the cost of road and road-rail transportation and the CO_2e emissions in both modes of transportation. We considered three road transportation routes (R1, R2, R3) in transportation based on the length of the trip. We used real-life data from 12 dry ports in Pakistan as this is the first study undertaken

Table 10. Computational Performance of Multi-Objective Genetic Algorithm (MOGA) in the Optimization of Bi-Objective Problems (Number of Iterations: 200)

Runs	Cost minimization MOGA computation time (seconds)	Greenhouse gas emission minimization MOGA computation time (seconds)
1	41	40
2	42	41
3	40	39
4	41	39
5	46	38
6	46	38
7	47	42
8	47	41
9	41	42
10	41	42
11	43	38
12	45	39
13	43	39
14	43	42
15	43	41
16	43	42
17	43	38
18	44	41
19	41	38
20	41	38
21	44	39
22	45	40
23	46	40
24	47	42
25	47	38

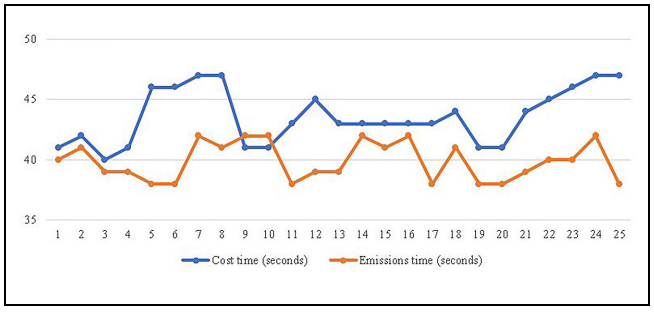


Figure 8. Computational time of multi-objective genetic algorithm (MOGA) (unit: seconds).

in Pakistan to consider the minimization of cost and CO_2e emissions as objectives. The Pareto-optimal solutions generated by the GA are provided to trade-off cost and CO_2e emissions. The results of the case study confirm that intermodal freight transportation is more economical and greener compared with road transportation.

Limitations and Future Work

In general, the suggested MILP model provides a valuable modeling framework for addressing cost and GHG emission concerns in Pakistan's transportation and logistics network. However, the research aims and objectives for road and intermodal transportation are defined too broadly in the MILP model, such as simple minimization of cost and GHG emissions. The formulations of costs and GHG emissions could be more precise (capital costs, line haulage costs, loading, off-loading, and transshipment costs and processes) and the study's focus could be expanded. Second, we did not take the time objective of transport between dry ports and seaports into account during the formulation and analysis of the problem. As a result, the study's weakness is its inability to evaluate transit time with regard to cost for each mode of transportation (cost of delays). Third, we focused exclusively on GHG emissions associated with green logistics. Other environmental concerns, such as noise, air pollution, and traffic accidents, are not addressed in this study and they may be addressed in future research. Additionally, our problem is solved using a MOGA, and CPU time could be measured using another evolutionary method to verify the algorithm's performance.

Author Contributions

The authors confirm contribution to the paper as follows: study conception and design: Z. Xiaoqiang; data collection: R. Shoukat; analysis and interpretation of results: R. Shoukat, Z. Xiaoqiang; draft manuscript preparation: R. Shoukat. All authors reviewed the results and approved the final version of the manuscript.

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The paper includes the data used to support the findings of the study

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